Differences in Community Recognition of Autism Between Special Education and Clinicians: Insights from Public Health Surveillance in Arizona

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Background

- Public health definition of autism spectrum disorder (ASD) is operationalized in three ways[1,2]:
  1. An autism special education exceptionality
  2. A documented ASD International Classification of Diseases (ICD) code
  3. A diagnostic statement from a clinical visit
- Rationale for this approach is based on the idea that neither education nor clinical sources alone give a complete picture of autism prevalence[2]
- Education and clinical definitions of ASD differ from each other
  - IDEA
  - DSM 5 diagnostic criteria
- Recognition from one source does not guarantee services provided by the other[3,4]
  - Whole subsets of children may not be able to access services

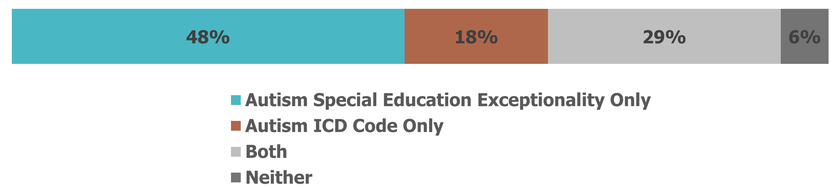
Objectives

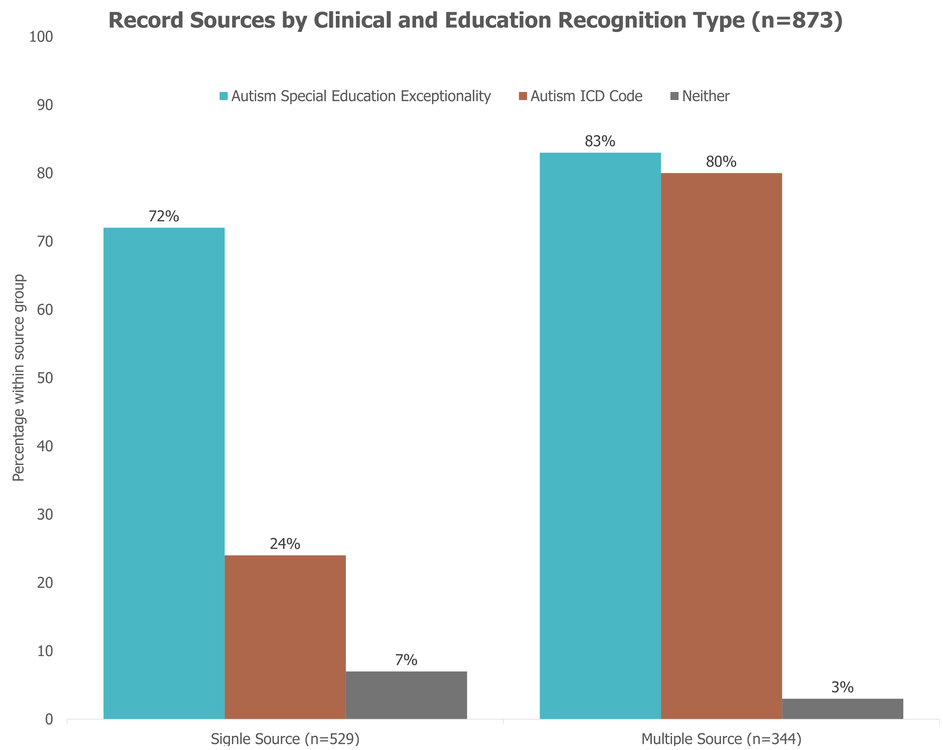
1. **Assess whether record source availability was associated differences in education and clinical recognition of ASD**
2. **Evaluate agreement between special education exceptionality and ICD coding for autism**
3. **Examine sociodemographic, intellectual disability, and record source factors in predicting education and clinical recognition of ASD**

Methods

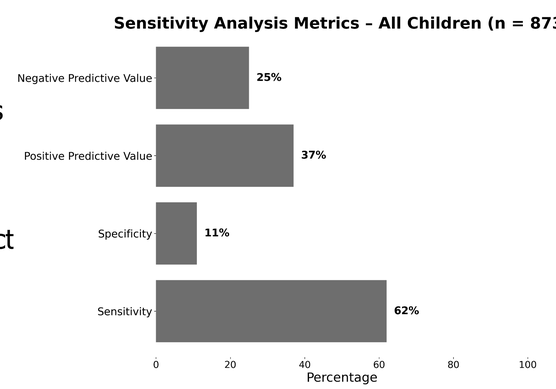
- Public health surveillance data from Arizona site of the ADDM Network on 8-year-olds collected in 2018, 2020, & 2022[1,5,6]
  - n = 873
- Variables included:
  - Demographics (race, ethnicity, sex)
  - Intellectual disability status
  - Surveillance year
  - Overall Social Vulnerability Index (SVI) Status
  - Record source
  - Education and clinical recognition of autism
- Data analysis:
  - Associations were assessed with Fisher's exact test
  - Cohen's κ
  - Sensitivity analyses
  - Penalized multinomial regression

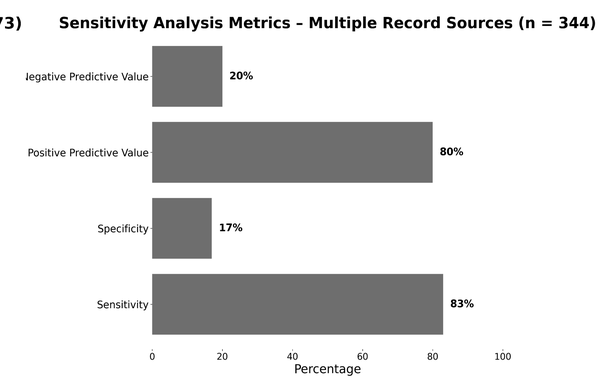
Results





*-Child significantly **more** likely to have autism special education exceptionality if they have multiple sources (OR=1.77, Holm p=0.002)*
*-Child significantly **more** likely to have an autism ICD code if they have multiple sources (OR=12.57, Holm p<0.001)*
*-Child significantly **less** likely to have neither special education or ICD code recognition for autism if they have multiple sources (OR=0.45, Holm p=0.018)*
*-Of children who have multiple sources, **Cohen's κ=-0.001***





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Implications

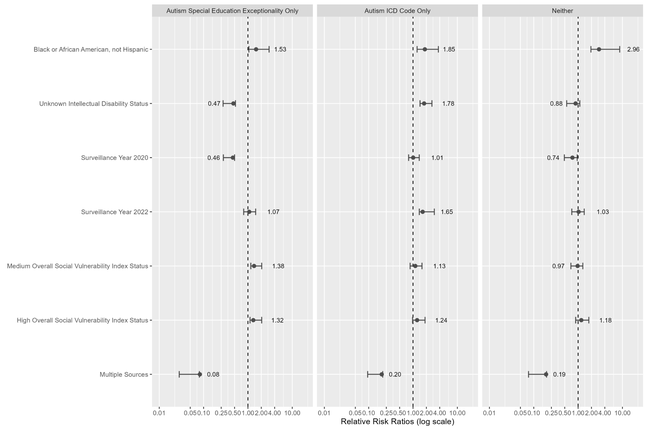

Figure 1. Statistically significant variables in a multinomial regression predicting membership in education and clinical recognition categories(n=873)

- Having multiple record sources **increases** likelihood of recognition from both education and clinical systems
- Other characteristics **decrease** likelihood of recognition from both education and clinical systems
- Are results a reflection of a **fundamental access issue** or are we seeing **two distinct subpopulations** of children with ASD?
- Need to explore whether children with special education exceptionality reliably qualify for a clinical ASD diagnosis

Conclusions

- Educational and clinical systems identify **overlapping but distinct** subsets of children with ASD
  - Special education exceptionality and ICD code for autism do not mutually predict each other
  - Suggests **system-level discordance**
  - Consistent with idea that special education requirements and clinical diagnostic criteria focus on different aspects of ASD
- May need to **reassess** how public health surveillance for autism is done
  - Are educational and clinical recognition of ASD the same thing?
  - Should diagnostic criteria be revised to include both educational and clinical experiences?
  - Findings need full ADDM Network confirmation

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